

UNFORGED · MIDNIGHT BUILDATHON · WAVE 2

Proof a photo is real. *Not proof of who took it.*

Anonymous, hardware-signed photo seals. A phone's secure chip signs the photo; a zero-knowledge proof on Midnight shows it came from one of a pool's enrolled phones, never which one.

github.com/bunnyyxtan/unforged

THE PROBLEM

Fakes are free. Proof of real *exposes people.*

$893M

2025 US losses in FBI complaints with an AI nexus

1 in 5

biometric fraud attempts are deepfakes (Entrust 2026)

17+

cameras and phones now sign photos at capture (C2PA, Pixel 10)

Signed capture proves a photo is real, but its metadata can reveal the photographer, the device and the location. For a reporter in a conflict zone, that's a physical danger.

Today you choose: **authentic and traceable**, or **anonymous and unverifiable**.

Sources: FBI IC3 2025; Entrust 2026 Identity Fraud Report; C2PA adoption trackers 2026; SMPTE 2025 "Privacy vulnerabilities in C2PA"; WITNESS. Links in docs/RESEARCH.md.

WHO NEEDS IT

The people whose photos matter most.

Stringers & citizen journalists

Need to be believed without being identified. Their safety depends on it.

Fact-checkers & photo desks

Need a one-second answer: real capture, unedited? Without learning the source.

Platforms, insurers, marketplaces

Must label real vs synthetic (EU AI Act Art. 50 since Aug 2026; India IT Rules since Feb 2026) without hoarding personal data.

Regulatory tailwind

EU AI Act Article 50 transparency duties apply from 2 Aug 2026. India's 2026 IT Rules amendment asks platforms to label synthetic media and embed provenance.

Our first design partners

Indian fact-checking organisations (IFCN signatories), then press-freedom NGOs and newsrooms with stringer networks.

THE SOLUTION

A seal for human-captured media.

1

Enroll

A phone creates a passkey in its secure chip (Secure Enclave, StrongBox, TPM). The pool stores only an opaque commitment in a Merkle tree.

2

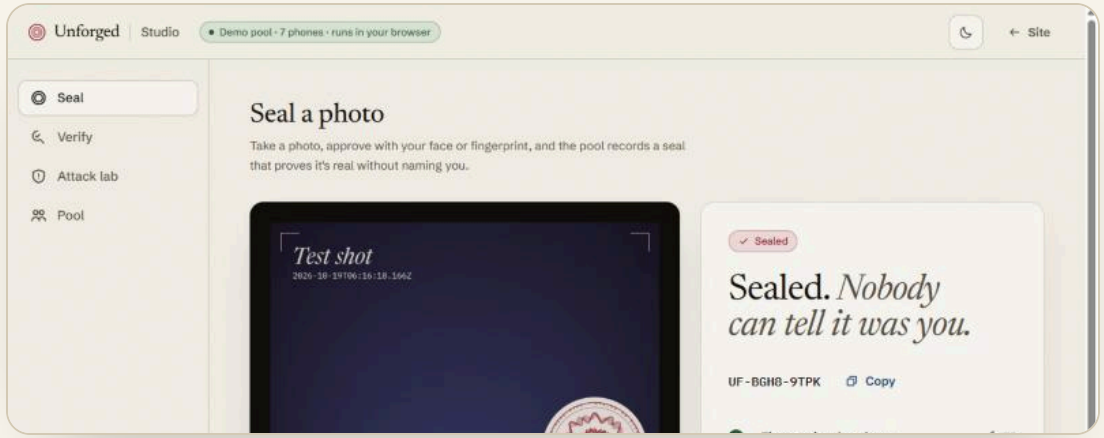
Seal

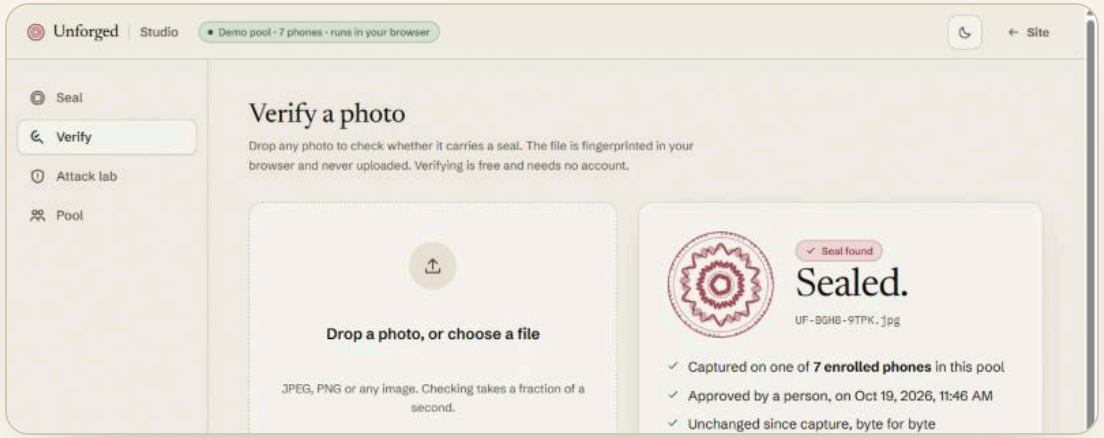
Face ID approves; the chip signs H(pool, SHA-256(photo)). A ZK proof shows the signer is *one of* the enrolled phones.

3

Verify

Anyone drops the photo: **Sealed**, with time and pool size. Photographer: hidden. Nobody can know.





BUILT ON MIDNIGHT

The chip's signature is verified *inside* the proof.

unforged.compact · Compact 0.35 · ZKIR v3

CIRCUIT	GUARANTEES
`enroll`	Admin adds an opaque device commitment to a HistoricMerkleTree
`seal`	Anonymous membership + in-circuit WebAuthn `secp256r1EcdsaVerify` over this pool + this photo, UP+UV, pool ≥ minimum, 24 h window
`revoke`	Removes a phone and resets accepted roots

Privacy boundary

PUBLIC ON MIDNIGHT	NEVER LEAVES THE PHONE
Photo fingerprint	Which phone sealed it
Capture time	Device key and salt
Pool size, epoch	Merkle path
Opaque commitments	Passkey signature, authenticator data

WebAuthn profile reused from Midnight Passport (Apache-2.0): clientDataJSON rebuilt in-circuit, RP and origin bound. Midnight's founder publicly floated a "ZK proof a photo was taken by a real Apple phone"; this is it, generalised to any passkey.

WHY ONLY MIDNIGHT

Authentic, anonymous, and checkable by anyone.

	CENTRAL DATABASE	PUBLIC CHAIN	C2PA ALONE	MIDNIGHT
Proves real capture	**trust vendor**	**yes**	**yes**	**yes**
Hides the photographer	**operator sees all**	**key on-chain**	**metadata leaks**	**ZK membership**
Verifiable without trust	**no**	**yes**	trust list	**yes**
Selective disclosure	**manual**	**no**	**no**	**native**
No crypto for users	**yes**	**wallet + gas**	**yes**	**sponsored DUST**

PROOF IT WORKS

Every attack is refused by the circuit.

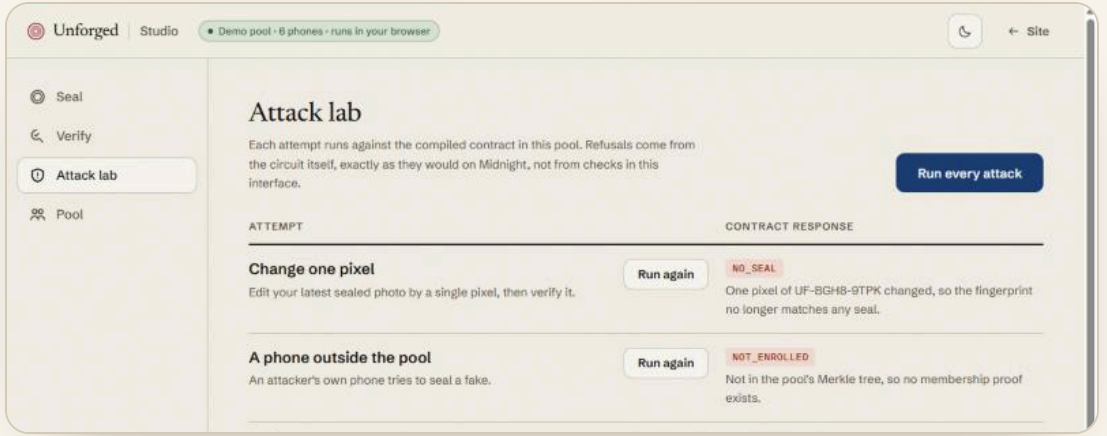
ATTACK	RESULT
Phone outside the pool	NOT_ENROLLED
Borrowed membership path	NOT_ENROLLED
Edited photo, reused signature	BAD_SIGNATURE
Replay from another pool	BAD_SIGNATURE
Stolen public key, no chip	BAD_SIGNATURE
Phishing origin / other website	REFUSED
No Face ID / PIN (UV)	REFUSED
Same photo twice	ALREADY_SEALED
Backdated / future photo	BAD_TIME
Pool too small to hide anyone	POOL_TOO_SMALL
Revoked phone	NOT_ENROLLED

26

tests, real P-256 passkey assertions

0

bytes differ in public state, whichever phone sealed



BUSINESS MODEL

Verification is free. Sealing is the product.

Verify

Free forever for anyone. API free tier. The public good that drives adoption.

Pools

Newsrooms & NGOs pay per enrolled device / month (hypothesis $8–15). Independent journalists ride NGO-sponsored pools free.

Capture SDK

Insurers, marketplaces, KYC: per sealed capture (hypothesis $0.05–0.50).

Verify API

Platforms labelling real vs synthetic at scale.

Moat: an anonymity network effect

A bigger pool hides everyone better, so shared pools get more valuable with every organisation that joins. Like Tor, for photographers.

Competition

Truepic, ProofMode, Numbers ProofSnap, OpenOrigins and native C2PA prove authenticity. None hides the signer while staying publicly verifiable. We are the privacy layer for content credentials.

ROADMAP

From seal to the trust layer for real media.

Wave 2 · shipped

Contract with in-circuit P-256 + anonymous membership. 26 tests. Web app with real passkeys, verify, attack lab, pool desk. Founding docs.

Wave 3

Ledger-9 network deploy (Preprod after the fork), phone PWA, Desk v1, sponsored fees, submission relay, 1–3 design partners sealing real photos.

Wave 4

Native iOS/Android with App Attest / Key Attestation (camera-bound pixels), device-bound keys, paid attribution ("claim this seal").

2027

C2PA bridge (Pixel, Leica → anonymous seal), coarse-location disclosure, edit-tolerant seals, video, Verify API and insurer pilots.

Honest limits today

A web passkey proves a person approved on an enrolled authenticator, not that the camera sensor produced the pixels; native attestation closes this in Wave 4. Anonymity is as strong as the pool is large; the contract enforces a minimum.

UNFORGED

Seal what's real.
Protect who saw it.

github.com/bunnyyxtan/unforged | Built on Midnight | Apache-2.0